

# 3D TransUNet for Multi-Class Brain Tumor Segmentation





**PROJECT TITLE :**

**AI-DRIVEN MULTI CLASS BRAIN TUMOR  
SEGMENTATION**

**DOMAIN :**

**MEDICAL IMAGE PREPROCESSING , DEEP LEARNING**

# **PROBLEM STATEMENT :**

- **Brain tumor segmentation helps in diagnosis, treatment planning, and patient monitoring.**
- **Manual MRI segmentation takes a lot of time and depends on expert doctors.**
- **Traditional CNN models cannot fully understand the complete brain context.**
- **Single-class segmentation is not enough for medical needs.**
- **An automated multi-class segmentation system is needed for accurate results.**
- **This system helps doctors and hospitals by providing fast and reliable tumor segmentation.**

# **PROJECT OBJECTIVES :**

- **To develop an automated deep learning model for multi-class brain tumor segmentation using MRI images.**
- **To combine CNN and Transformer models to capture both local and global features.**
- **To accurately identify different tumor regions such as edema, tumor core, and enhancing tumor.**
- **To improve segmentation accuracy compared to CNN-only models.**
- **To support doctors by providing fast, consistent, and reliable automated tumor segmentation.**

# **DATASET DESCRIPTION**

- **Dataset Name: Brain Tumor Segmentation (BraTS) 2021**
- **Type: Publicly available medical imaging dataset**
- **Application: Brain tumor detection and segmentation**

## **MRI MODALITIES**

- **T1: Shows normal brain anatomy**
- **T1CE (Contrast Enhanced): Highlights active tumor regions**
- **T2: Visualizes fluid and swelling areas**
- **FLAIR: Clearly shows edema around the tumor**

## **SEGMENTATION CLASS**

- **Background: Normal brain tissue**
- **Edema: Swollen tissue surrounding the tumor**
- **Tumor Core: Main tumor mass**
- **Enhancing Tumor: Actively growing tumor region**

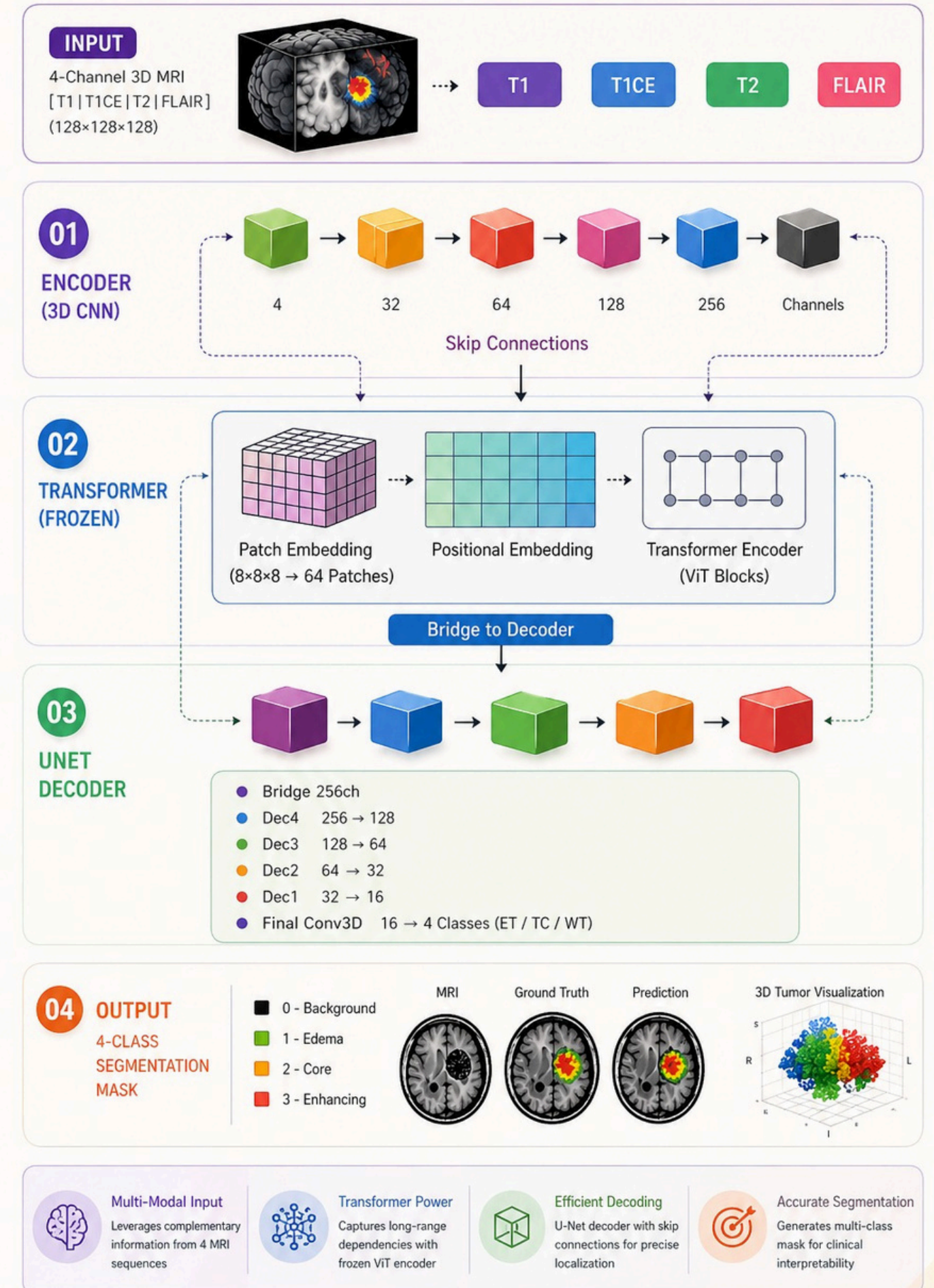


# TRANSUNET LAYERS

- CNN Encoder Layer – finds important details in the image
- Patch Embedding Layer – splits the image into small parts
- Positional Embedding Layer – remembers where each part is
- Transformer Encoder Layer – understands the whole image
- Transformer–CNN Bridge – combines all learned features
- Decoder Layer – rebuilds the image
- Segmentation Head – shows the final tumor area

## BRAIN TUMOR SEGMENTATION

DEEP LEARNING PIPELINE



# VALIDATION OUTPUT

| Class      | Dice Score | IoU    | Sensitivity | Specificity |
|------------|------------|--------|-------------|-------------|
| Background | 0.9908     | 0.9819 | 0.9866      | 0.7651      |
| NCR/NET    | 0.3116     | 0.2315 | 0.2503      | 0.9864      |
| ED         | 0.5799     | 0.4471 | 0.825       | 0.9964      |
| ET         | 0.4887     | 0.4032 | 0.5537      | 0.9986      |
| TUMOR MEAN | 0.46       | 0.3606 | 0.543       | 0.9938      |

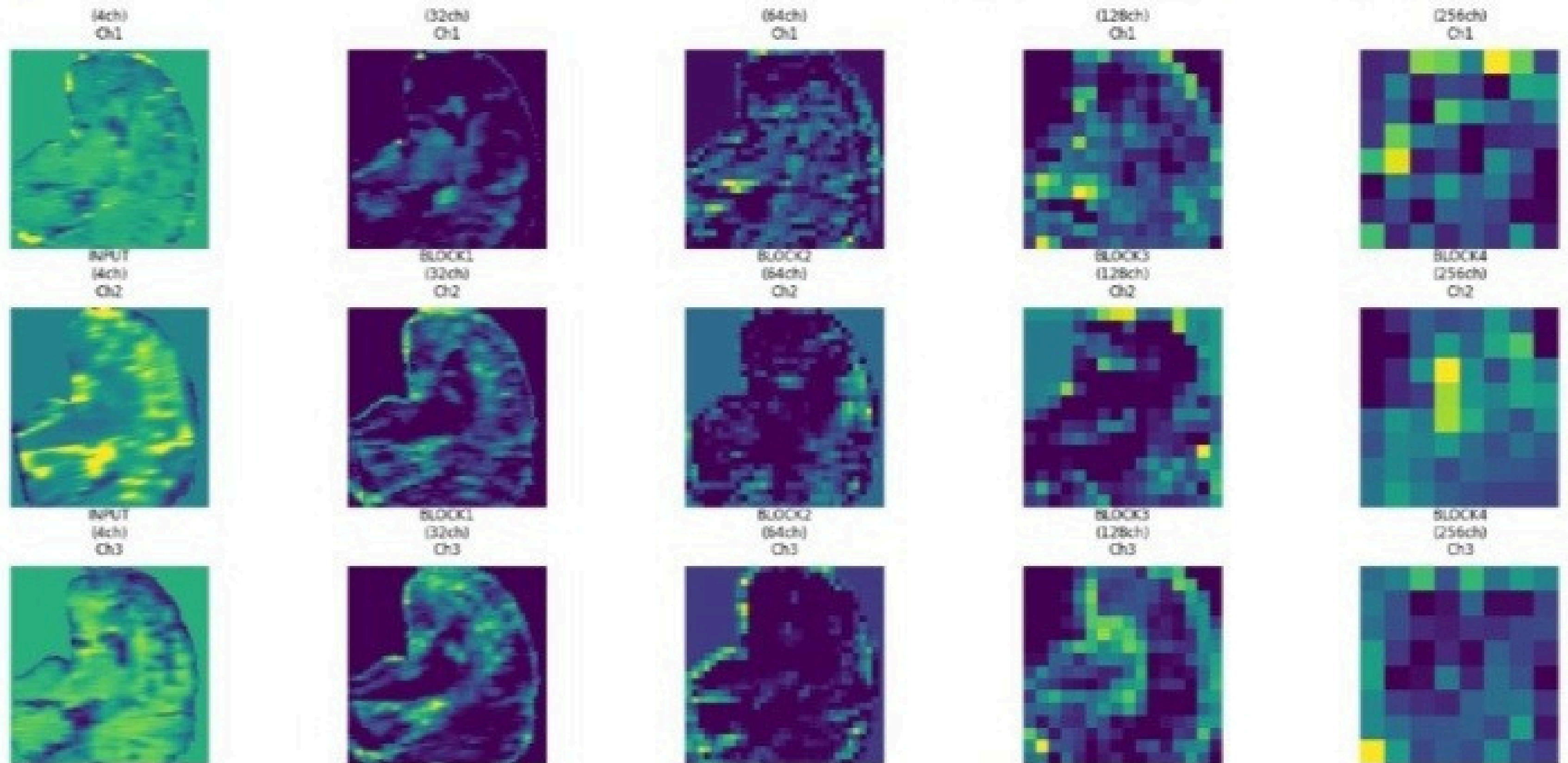


# MATRICES

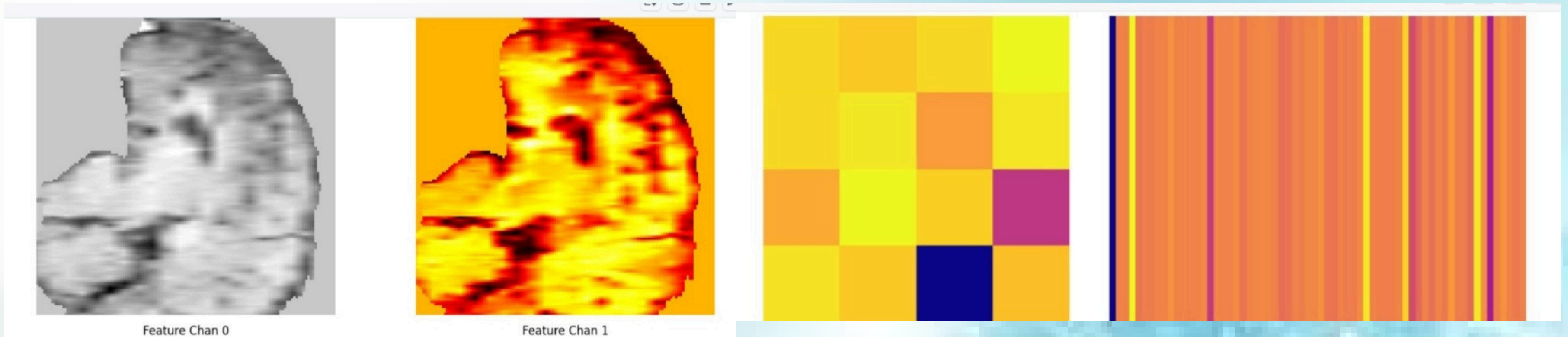
| Metrics                       | Formula                                                                                              |                                   |                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------------------|
| 95% Hausdorff distance (HD)   | $95\% \text{ HD} = P_{95} \left( \min_{a \in A}   B - A  _2 \cup \min_{b \in B}   A - B  _2 \right)$ | Dice similarity coefficient (DSC) | $\text{DSC} = \frac{2 \times \text{IoU}}{1 + \text{IoU}}$    |
|                               |                                                                                                      | Precision                         | $\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$ |
| Intersection over union (IoU) | $\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}$                                   | Recall                            | $\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$    |



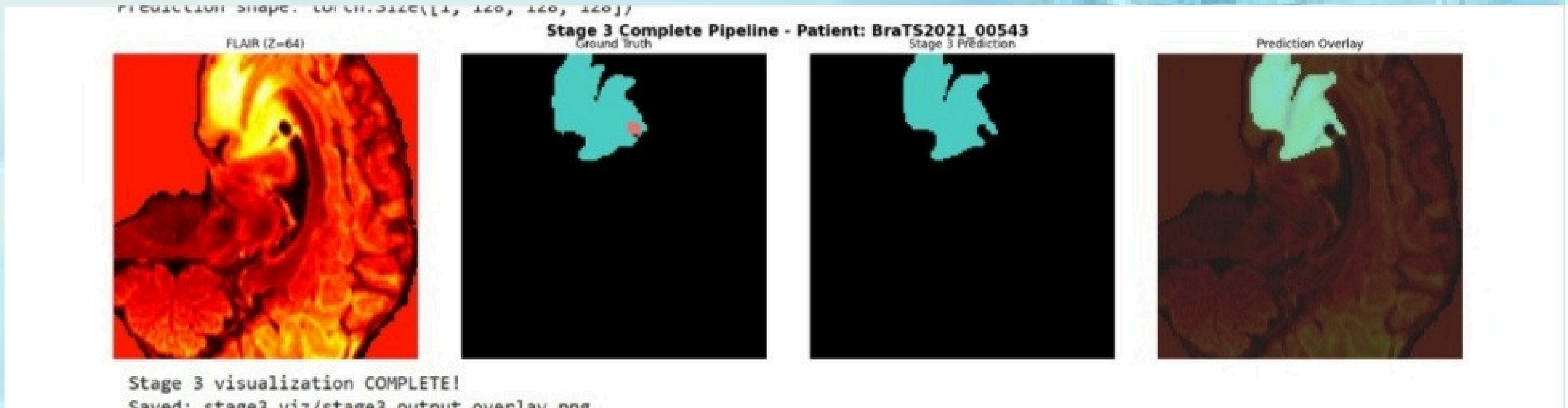
# OUTPUT STAGE 1



# OUTPUT STAGE 2



# OUTPUT STAGE 3





# TESTING OUTPUT PER PATIENT

TESTING PATIENT: BraTS2021\_01565

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## SEGMENTATION PERFORMANCE:

Overall Accuracy: 99.4%  
Dice: 0.9228  
IoU: 0.8628  
Sensitivity: 0.9700  
Specificity: 0.9990  
HD95: 0.08mm

## TUMOR COMPOSITION:

Non-ET (NCR/NET): 0.55%  
Edema: 4.69%  
ET (Enhancing): 1.28%  
Total Tumor: 6.51%

CORRECT: TUMOR DETECTED ✓

Types: Non-ET, Edema, ET  
Dominant: Non-ET

OVERALL ACCURACY: 99.4%

# TESTING OVERALL

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## FINAL RESULTS - ALL 126 PATIENTS

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## AVERAGE SEGMENTATION PERFORMANCE:

Accuracy : 99.217  
Dice : 0.837  
IoU : 0.778  
Sensitivity : 0.905  
Specificity : 0.998  
HD95\_mm : 0.075

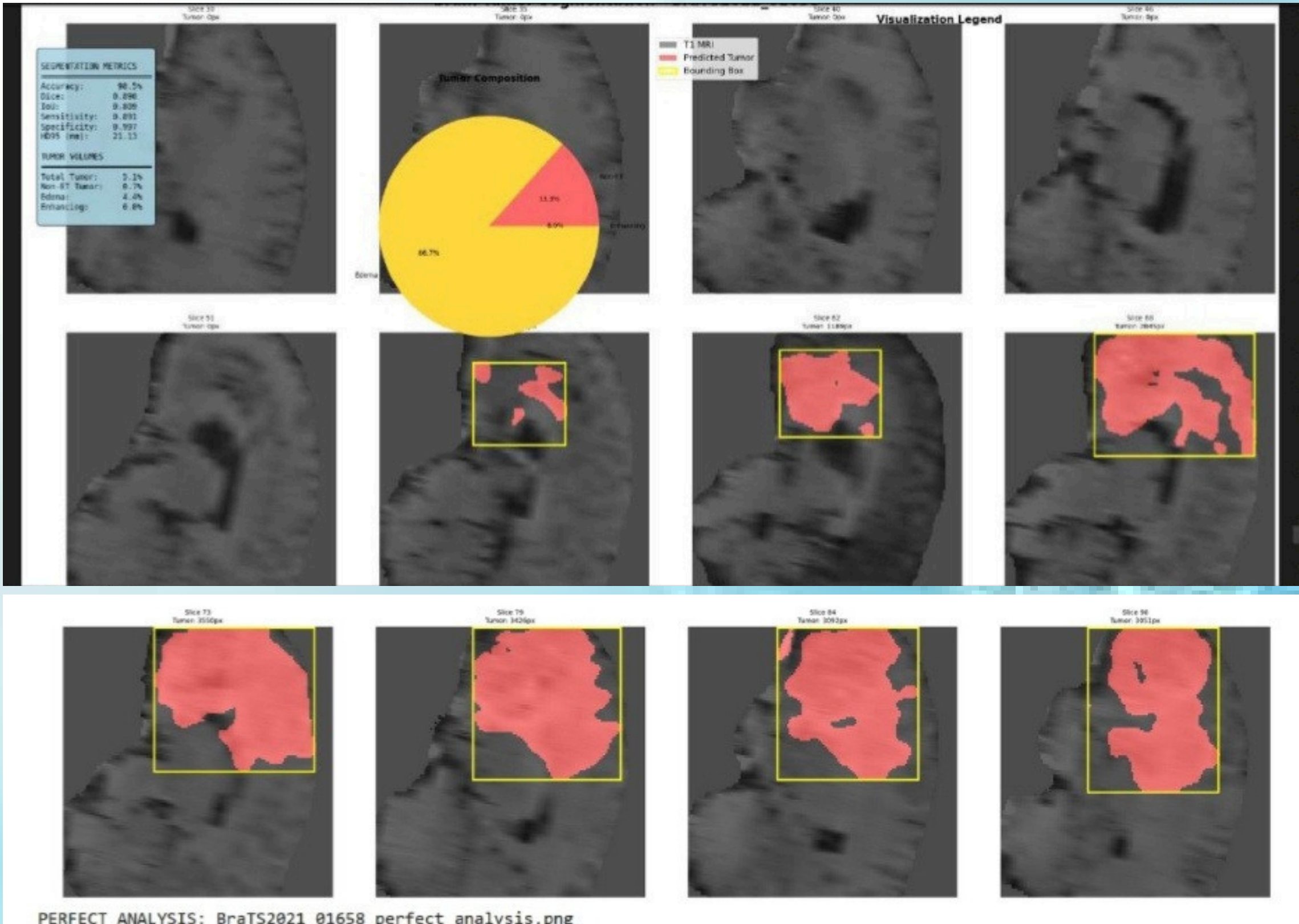
## AVERAGE TUMOR COMPOSITION:

Non-ET (NCR/NET): 0.45%  
Edema: 2.80%  
ET (Enhancing): 0.99%  
Total Tumor: 4.23%

TUMOR DETECTION RATE: 112/126 (88.9%)

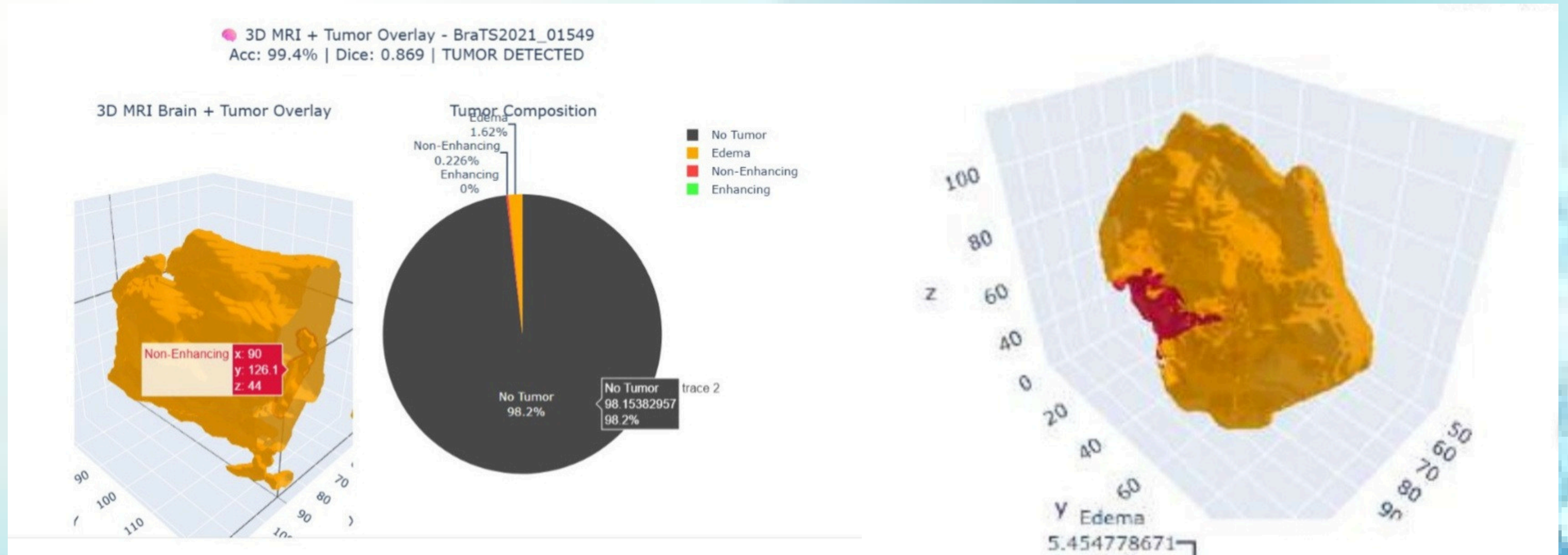
OVERALL ACCURACY: 99.2%

# TESTING OUTPUT 2D





# TESTING OUTPUT 3D



# APPLICATION AND IMPACTS

- **Brain tumor diagnosis and treatment planning.**
- **Surgical guidance and tumor localization.**
- **Radiotherapy planning.**
- **Medical research and analysis.**
- **Reduces workload of radiologists and clinicians.**
- **Improves accuracy and efficiency in healthcare systems.**



# CHALLENGES

- Dealing with incorrect or missing labels in segmentation masks
- Managing large 3D MRI data with limited system memory
- Class imbalance between tumor and background regions
- Long training time and performance optimization issues

# LEARNING AND SKILLS GAINED

- Understanding medical MRI images and tumor regions
- Working with medical image formats (NIfTI)
- Applying deep learning models for image segmentation
- Using CNN and Transformer architectures together
- Evaluating models using Dice, IoU, Sensitivity

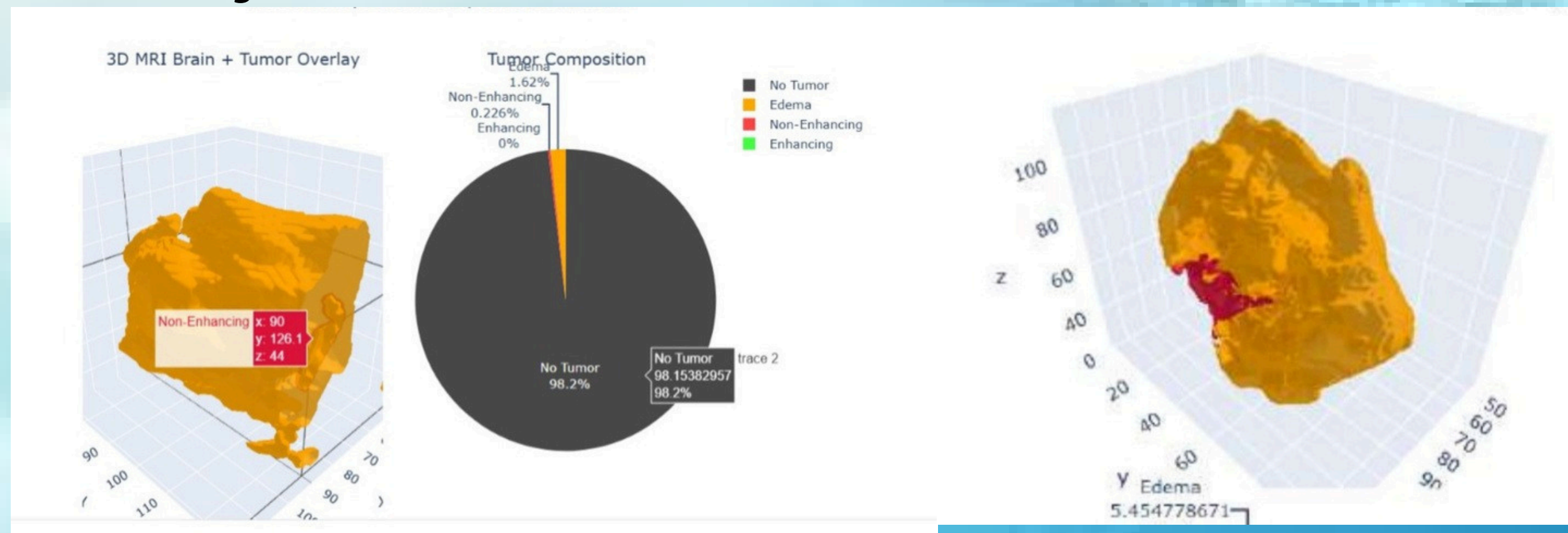
# FUTURE WORK

- **Deploy the model as a web or desktop application for real-time use**
- **Implement Vision Transformer (ViT) and compare results with TransUNet**
- **Train on larger and more diverse datasets for better robustness**
- **Optimize model performance for faster inference and lower memory usage**
- **Clinical validation with expert-reviewed annotations**



# CONCLUSION

- A TransUNet-based model was developed for multi-class brain
- tumor segmentation.
- The model effectively combines CNN and Transformer
- advantages.
- Achieved accurate and reliable segmentation results.
- Demonstrates strong potential for clinical applications.
- Automated segmentation can significantly improve healthcare
- efficiency.



*Thank You*